

# A Practical Guide to Ranking Data Analysis in the Social Sciences

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# Examples in Political Sciences

| Study | Area | Predictor | Ranking Outcome |
|-------|------|-----------|-----------------|
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| Atsusaka (2025) | Method | ballot order    | candidate rankings   |

# Practical Ranking Data Analysis

- Statistical computing
  - **rdy**: <https://github.com/YukiAtsusaka/rdy>
  - **rankingQ**: <https://sysilviakim.com/rankingQ>

1. Data  
collection

2.  
Descriptive  
statistics

3. Rank  
distances

4. Modeling

5. Drawing  
quantities of  
interest

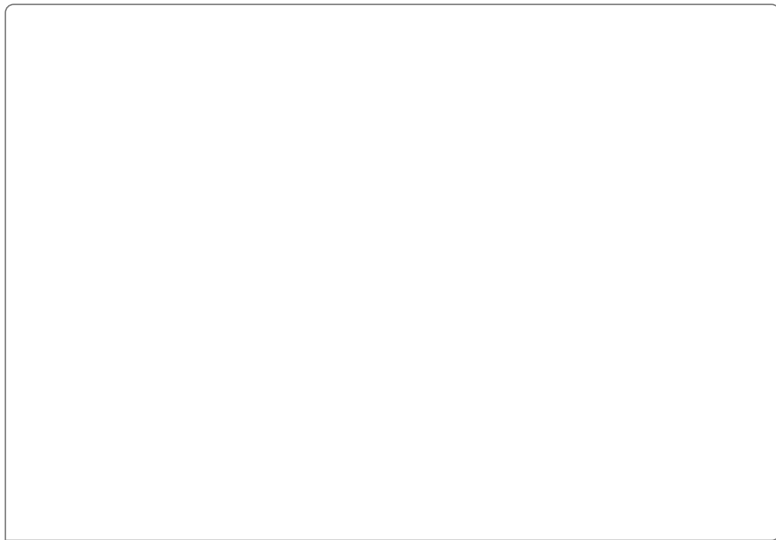
6.  
Sampling/  
simulations

7. Power  
calculation

8. Missing  
data

9.  
Extensions

# Motivating Application





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↪ Black and white Americans have different ranked preferences for representation (Atsusaka & Hayes, 2025)

## Motivating Application

|   | ranking | policy | service | allocation | contact | trust | race |
|---|---------|--------|---------|------------|---------|-------|------|
| 1 | 24351   | 2      | 4       | 3          | 5       | 1     | 1    |
| 2 | 14235   | 1      | 4       | 2          | 3       | 5     | 1    |
| 3 | 25413   | 2      | 5       | 4          | 1       | 3     | 1    |
| 4 | 21435   | 2      | 1       | 4          | 3       | 5     | 3    |
| 5 | 15324   | 1      | 5       | 3          | 2       | 4     | 1    |
| 6 | 25314   | 2      | 5       | 3          | 1       | 4     | 1    |



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    - ↪ Bias correction (plug-in/inverse probability weighting)

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In American politics, there are multiple levels of government, where elected officials can represent their constituents. Please rank the following five levels of government from the highest to the lowest. The level of government is higher when it covers a broader range of government.

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  - ↪ A lot of quantities of interest!



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| ranking | ordering                                        | %   |
|---------|-------------------------------------------------|-----|
| 13245   | policy > allocation > service > contact > trust | 5.4 |
| 14235   | policy > allocation > contact > service > trust | 5.4 |
| 14325   | policy > contact > allocation > service > trust | 4.2 |
| 15234   | policy > allocation > service > trust > service | 3.4 |

Table: Modal Rankings among Americans

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(Equivalent of univariate analysis)

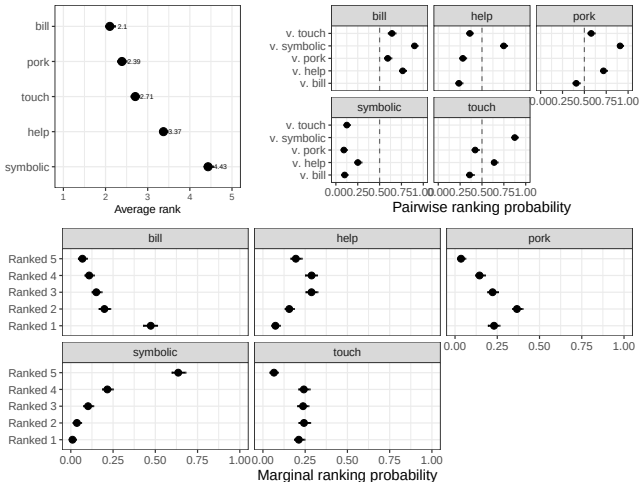
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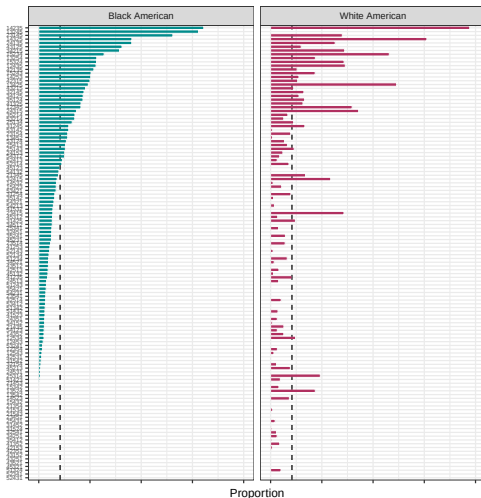
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rankingQ::impr_r_weight(data = rep, "c_rep", "anchor")
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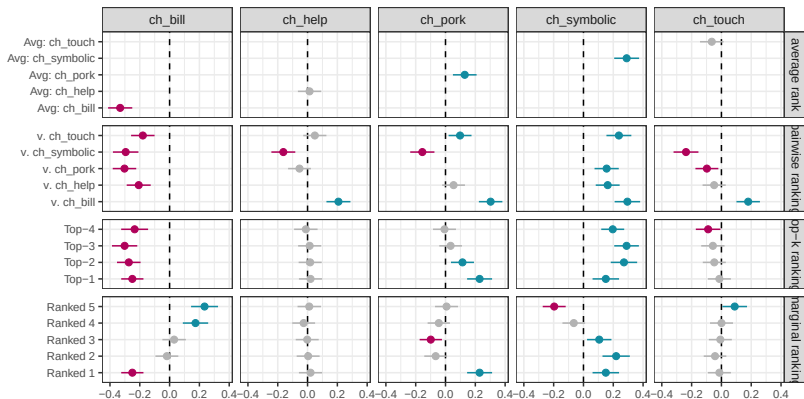
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```
rdy::diff_in_mean(dt, item_names, "black") %>%  
  rdy::viz_diff(scale = "std")
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rdy::rank_dist(dt, group_col = "black",  
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```

| Method                                   | $d_{\max}$ | $\hat{\theta}$ (0–1) | SE    | 95% CI       |
|------------------------------------------|------------|----------------------|-------|--------------|
| Kendall $\tau$ (# pairwise disagreement) | 10         | 0.45                 | 0.003 | [0.44, 0.45] |
| Footrule (sum of absolute difference)    | 12         | 0.61                 | 0.004 | [0.60, 0.61] |
| Spearman's $\rho$ (correlation)          | 40         | 0.43                 | 0.004 | [0.43, 0.44] |

**Table:** Difference between Rankings by Black and White Americans over Five Items ( $J = 5$ ,  $n_W = 1388$ ,  $n_B = 1201$ ).

## 4. Modeling via the Plackett-Luce Model

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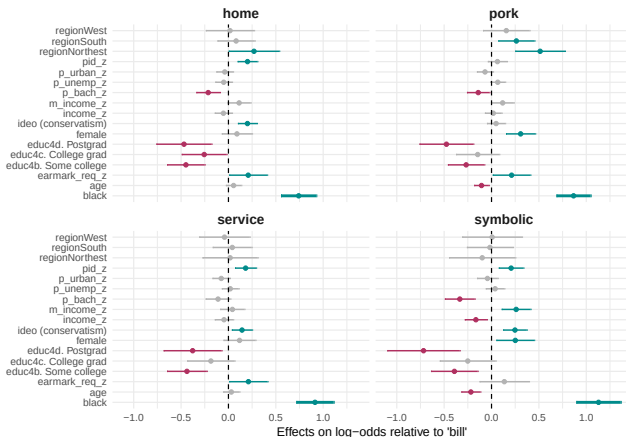
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```
mlogit::mlogit(ch ~ 1 | female + age + educ + ideology +  
pid + income + region, data = mdat, weight = dual_weight,  
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→ rdy-function forthcoming

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prob = c(0.4, 0.3, 0.2, 0.1),  
choices = c("safe", "perform", "revenue", "biling"))
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|   | 1st     | 2nd     | 3rd     | 4th     |
|---|---------|---------|---------|---------|
| 1 | safe    | perform | biling  | revenue |
| 2 | perform | revenue | biling  | safe    |
| 3 | biling  | revenue | safe    | perform |
| 4 | safe    | perform | revenue | biling  |
| 5 | safe    | biling  | perform | revenue |
| 6 | perform | safe    | biling  | revenue |

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| id | ch_safe | ch_perform | ch_revenue | ch_biling | age | kids |
|----|---------|------------|------------|-----------|-----|------|
| 1  | 1       | 4          | 2          | 3         | 36  | 0    |
| 2  | 2       | 4          | 1          | 3         | 38  | 0    |
| 3  | 1       | 4          | 2          | 3         | 25  | 1    |
| 4  | 2       | 4          | 1          | 3         | 45  | 1    |
| 5  | 1       | 4          | 2          | 3         | 26  | 1    |
| 6  | 1       | 4          | 2          | 3         | 34  | 1    |

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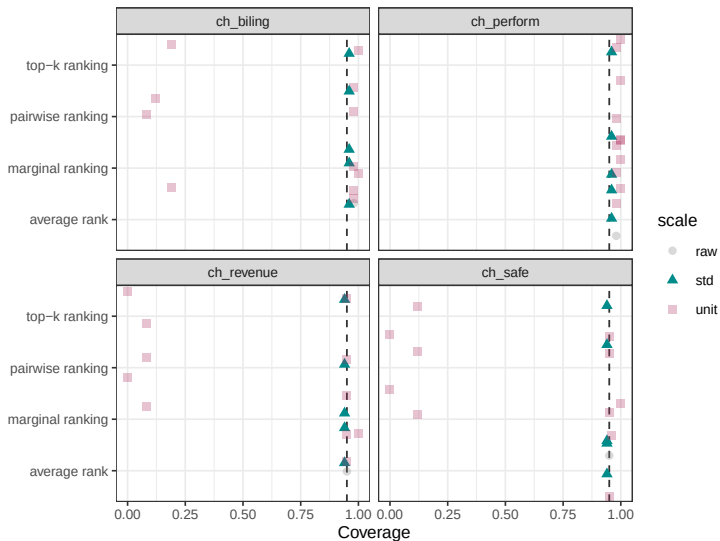
| id | treat | ch_safe | ch_perform | ch_revenue | ch_biling | age | kids |
|----|-------|---------|------------|------------|-----------|-----|------|
| 1  | 0     | 1       | 4          | 2          | 3         | 36  | 0    |
| 2  | 1     | 2       | 4          | 1          | 3         | 38  | 0    |
| 3  | 0     | 1       | 4          | 2          | 3         | 25  | 1    |
| 4  | 1     | 2       | 4          | 1          | 3         | 45  | 1    |
| 5  | 1     | 1       | 4          | 2          | 3         | 26  | 1    |
| 6  | 1     | 1       | 4          | 2          | 3         | 34  | 1    |

## 7. Inference and Power Calculation

Robust inference with standardized differences ( $\tau_{\text{ATE}}/sd(Y_i|D_i = 0)$ )

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Robust inference with standardized differences ( $\tau_{\text{ATE}}/sd(Y_i|D_i = 0)$ )



## 7. Inference and Power Calculation

Simulation-based power analysis

## 7. Inference and Power Calculation

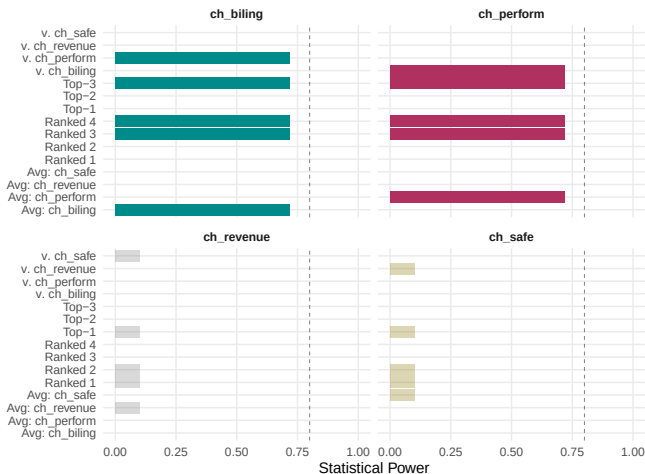
### Simulation-based power analysis

```
rdy::sim_power(pop, n_sample = 500, n_rep = 50)
```

# 7. Inference and Power Calculation

## Simulation-based power analysis

```
rdy::sim_power(pop, n_sample = 500, n_rep = 50)
```



## 7. Inference and Power Calculation

### The Problem of Degenerated Outcomes in the Control

- All control units have the same preference with respect to a given item
  - $\text{Var}(Y_{\text{obs}}|D=0) = 0$
  - No information about the distribution (effect size is unknown)
  - Standardized diff-in-means will be undefined
- What degeneracy may imply
  - No heterogeneity in the preference among controls
  - The sample is too small to spot variation
  - Potential floor/ceiling effects

## 8. Missing Data



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  - One or more items are not ranked

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| Level     | Variable | Missing |
|-----------|----------|---------|
| Candidate | Peltola  | 0.41    |
| Candidate | Bye      | 0.64    |
| Candidate | Begich   | 0.35    |
| Candidate | Palin    | 0.43    |
| Rank      | rank1    | 0.00    |
| Rank      | rank2    | 0.33    |
| Rank      | rank3    | 0.67    |
| Rank      | rank4    | 0.81    |

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